# SOLUTIONS

# Practice Paper–05

## SECTION A

**1.** (b) Destructive interference

**Why?** A phase difference of 180° means the waves are completely out of phase and will cancel each other out, leading to destructive interference.

**Concept:** Interference

**2.** (b) Copper

**Why?** Copper has all paired electrons which is property of Diamagnetic material.

**Concept:** Diamagnetic material

**3.** (b) The angle between the coil and the magnetic field is 90°

**Why?** The maximum induced emf in an AC generator occurs when the angle between the coil and the magnetic field is 90°, maximizing the rate of change of flux.

**Concept:** Induced emf

**4.** (a) $1.6 \times 10^{-17}$ J

**Why?** The kinetic energy KE gained by an electron when accelerated through a potential difference $V$ is given by:

$$KE = eV$$

where $e = 1.6 \times 10^{-19}$ C and $V = 100$ V.

Therefore:

$$KE = 1.6 \times 10^{-19} \times 100 = 1.6 \times 10^{-17} \text{ J}$$

**Concept:** Kinetic Energy of electron

**5.** (b) Inversely proportional to their radii

**Why?** In this case, the potential V is the same for two spheres, then

$$\frac{kQ_1}{R_1} = \frac{kQ_2}{R_2}$$

$$\frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

So, $Q \propto R$

i.e., $Q = kR$

Substituting this relationship into the surface charge density formula:

$$\sigma = \frac{Q}{4\pi R^2} = \frac{kR}{4\pi R^2} = \frac{k}{4\pi R}$$

This shows that the surface charge density (σ) is inversely proportional to the radius (R) of the sphere, $\sigma \propto \frac{1}{R}$.

**Concept:** Electric potential

**6.** (b) Particle nature of light

**Why?** The photoelectric effect demonstrates that light behaves as a particle, where energy is transferred in discrete packets (photons).

**Concept:** Photoelectric effect

**7.** (a) Negative

**Why?** The total mechanical energy of an electron in an atom is negative because the electron is bound to the nucleus, indicating a stable orbit.

**Concept:** Nucleus

**8.** (a) Ferromagnetic materials

**Why?** Magnetic susceptibility indicates how much a material is magnetized in a magnetic field. Ferromagnetic materials have a large positive magnetic susceptibility, indicating a strong attraction to magnetic fields, unlike paramagnetic (small positive susceptibility) and diamagnetic (small negative susceptibility) materials.

**Concept:** Magnetic material

**9.** (d) The electric field between the plates remains the same.

**Why?** When a dielectric slab is inserted into a charged capacitor still connected to a battery, the battery acts as a constant voltage source, ensuring the potential difference across the plates remains unchanged. The dielectric material itself becomes polarized within the electric field, inducing its own internal electric field that opposes the original field from the capacitor's charges. This effectively reduces the net electric field strength between the plates.

**Concept:** Capacitance

**10.** (c) They are transverse in nature.

**Why?** Electromagnetic waves are transverse waves, meaning the oscillations of the electric and magnetic fields are perpendicular to the direction of propagation. They do not require a medium for propagation and can travel through a vacuum.

**Concept:** EM Waves

**11.** (b) Knee voltage decreases, reverse leakage current increases.

**Why?** When temperature increases:

Knee voltage drops because the semiconductor's band gap gets smaller. Reverse leakage current rises because more charge carriers are thermally generated.

**Concept:** Semiconductor

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**12.** (b) To convert AC into DC

**Why?** A rectifier allows only one direction of current (because diode conducts only in forward bias).

**Concept:** Rectifier

**13.** (c) Assertion is true, but Reason is false.

**Why?** Electric field lines don't form closed loops because they originate from positive charges and end on negative charges. However, moving a charge along an electric field line generally involves non zero work

$W = q\int \vec{E}\cdot\vec{dl}$

**Concept:** Electric field

**14.** (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

**Why?** Reflecting telescopes use mirrors, which don't suffer from chromatic aberration (colour distortion), unlike the lenses in refracting telescopes. This makes them better for clear astronomical observations.

**Concept:** Telescope

**15.** (c) Assertion is true, but Reason is false.

**Why?** Photoelectric effect demonstrates light's particle nature. However, the Reason is false; the number of photoelectrons emitted is proportional to the photon flux $\left(\phi = \frac{I}{h\nu}\right)$ of light, not its frequency.

**Concept:** Photoelectric effect

**16.** (b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

**Why?** Kirchhoff's First Law states that the sum of currents at a junction is zero. This is a direct application of the principle that charge is conserved; But reason does not explain to the Assertion. It should be $\frac{dQ}{dt} = 0 \Rightarrow \sum I = 0$

**Concept:** Kirchhoff's law

## SECTION B

**17.** Doping increases semiconductor conductivity by adding impurities.

N-type (e.g., Silicon + Phosphorus) uses pentavalent dopants, adding excess electrons (majority carriers).

P-type (e.g., Silicon + Indium) uses trivalent dopants, creating excess holes (majority carriers).

**Concept:** Semiconductor

**18.** **(i)** Temperature Coefficient ($\alpha$):

Using $R_t = R_0\,(1 + \alpha(T - T_0))$:

$18 = 15[(1 + \alpha(100 - 20)]$

$18 = 15 + 15\alpha \times 80$

$3 = 1200\alpha$

$\alpha = 0.0025°C^{-1}$

**(ii)** Resistivity ($\rho$) at 20°C :

Diameter = 0.6 mm = $0.6 \times 10^{-3}$ m

Radius ($r$) = $0.3 \times 10^{-3}$ m

Area ($A$) = $\pi(0.3 \times 10^{-3})^2 \approx 0.28 \times 10^{-6}\ m^2$

Resistivity $\rho = \frac{R \times A}{L}$

$\rho = \frac{15\Omega \times 0.28 \times 10^{-6} m^2}{2\ m}$

$\rho \approx 2.1 \times 10^{-6}\ \Omega\ .\ m$

**Concept:** Resistivity

**19.** **(I) (a)** Wavelength $\lambda = \frac{c}{f}$

$\lambda = \frac{(3.0 \times 10^8\ ms^{-1})}{(3.0 \times 10^9\ Hz)}$

$\lambda = 1 \times 10^{-1}$ m = 0.1 m or 10 cm

**(b)** Again, the magnetic field is given by $B = \frac{E}{c}$

$B = (60\ Vm^{-1})\ (3.0 \times 10^8\ ms^{-1})$

$B = 20 \times 10^{-8}\ TB = 2 \times 10^{-7}$ T

**Concept:** EM Waves

**OR**

**(II) (i) Requirement of a Medium:** Electromagnetic waves do not require a material medium (solid, liquid or gas) for their propagation. They can travel through a vacuum. In contrast, mechanical waves require a medium to transfer energy. For instance, sound waves, a type of mechanical wave, need air, water or a solid to travel.

**(ii) Nature of the Wave:** Electromagnetic waves are transverse in nature. This means the oscillations of the electric and magnetic fields that constitute the wave are perpendicular to the direction of wave propagation. Mechanical waves can be both transverse or longitudinal. In a transverse mechanical wave, the particles of the medium oscillate perpendicular to the direction of wave propagation (e.g., waves on a string or water surface waves). In a longitudinal mechanical wave, the particles of the medium vibrate parallel to the direction of wave propagation (e.g., sound waves).

**Concept:** EM Waves

**20.** **(I)** $\frac{I_{max}}{I_{min}} = \frac{25}{16}$ or $\left(\frac{r+1}{r-1}\right)^2 = \frac{25}{16}$

$\frac{r+1}{r-1} = \frac{5}{4}$ or $5r - 5 = 4r + 4$

$r = 9 = \frac{a_1}{a_2}$, the amplitude ratio.

Width ratio of two slits,

$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{81}{1} = 81:1$

**Concept:** Interference of light.

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**Commonly Made Error:**

- Many students directly divide 25 by 16 and forget to take square root when applying formula.

**Answering Tips:**

- Recall correct relation

$$\frac{I_{max}}{I_{min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}$$

- Always take square root of intensity ratio before relating it to slit widths

$$\sqrt{\frac{I_{max}}{I_{min}}} = \frac{a_1 + a_2}{a_1 - a_2}$$

**OR**

**(II) (a)** Condition for constructive interference

$$x_n = (4n-1)\frac{\lambda D}{4d}$$

Condition for destructive interference

$$x_n = (4n+1)\frac{\lambda D}{4d}$$

**(b)** Fringe width, $\beta = x_{n+1} - x_n$

$$= \frac{\lambda D}{d}\left[(n+1-\frac{1}{4}-(n-\frac{1}{4})\right]$$

$$= \frac{\lambda D}{4}$$

**Concept:** Interference of light.

**21.** **(i)** in loop (1) - Anticlockwise in loop (2) – Clockwise

**(ii)** No, emf will not be equal because the rate of change of magnetic flux in the two loops is different.

**Concept:** Induced emf

## SECTION C

**22.** Let $r$ and $R$ be the radii of the small and bigger drops, respectively.

Volume of the bigger drop

$= 64 \times$ Volume of a small drop

i.e., $= \frac{4}{3}\pi R^3 = 64 \times \frac{4}{3}\pi r^3$

$R = 4r = 4 \times 9$ mm $= 36 \times 10^{-3}$ m

$\therefore$ Capacitance of the bigger drop is

$$C = 4\pi\varepsilon_0 R = \frac{1}{9\times10^9}\cdot 36\times10^{-3}\,\text{F}$$

$$= 4 \times 10^{-12}\,\text{F} = 4\,\text{pF}$$

Charge on bigger drop

$q = 64 \times$ Charge on a small drop

$= 64 \times 10^{-12}$ C

$\therefore$ Potential of bigger drop is

$$V = \frac{q}{C} = \frac{64\times10^{-12}}{10^{-12}} = 64\text{V}$$

**Concept:** Capacitance

**Commonly Made Errors:**

- Students multiply charge directly to find capacity
- Forget to make cube root of $n$ and instead use $R = nr$

**Answering Tips:**

- Capacity of spherical conductor depends only on its radius not on charge
- When many drops combine use volume conservation: $R = n^{\frac{1}{3}}r$

**23.** Electric field lines are imaginary curves drawn in an electric field to show the path a tiny, free positive charge would take. The direction of the electric field at any point on this path is given by the tangent to the curve at that point.

**Properties:**

**1. Continuous and Smooth:** These lines are unbroken paths. They do not have sudden jumps or sharp corners.

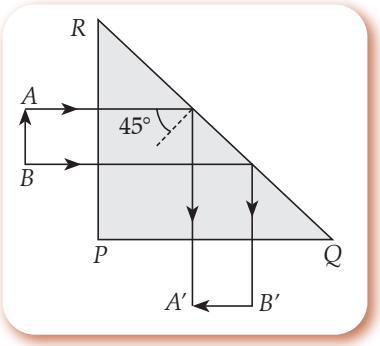
**2. Origin and Destination:** Electric field lines begin at positive charges and end at negative charges. They do not form closed loops. If only one charge exists, the lines will start or end far away.

**3. Direction at a Point:** A line that touches an electric field line at any point (a tangent) will point in the exact direction of the electric field at that location.

**4. No Intersections:** Electric field lines cannot cross each other. If they did, the electric field at the crossing point would have two directions at once, which is impossible.

**Concept:** Electric field

**24.** **(i)** A totally reflecting prism is a type of right-angled isosceles prism, meaning it has one angle of 90° and two other angles of 45° each.




**(ii)** When $i = i_c$, $r = 90°$

Using Snell's law of refraction, $\mu_1 \sin i_c = \mu_2 \sin 90°$

$$\sin i_c = \frac{\mu_2}{\mu_1}$$

$$i_c = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\right)$$

**Concept:** Prism

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**25.** **(i)** The condition for the first minimum in single-slit diffraction is given by: $a \sin\theta = n\lambda$

Where $n = 1$ for the first minimum.

For small angles, $\sin\theta \approx \theta$ (in radians).

So, for the first minimum, $a\theta = \lambda$ or $\theta = \lambda a$.

The angular width of the central maximum ($2\theta$) is twice the angle to the first minimum on either side of the center.

Therefore, $2\theta = \dfrac{2\lambda}{a}$

Substitute the given values,

$$2\theta = \frac{(2\times500\times10^{-9})}{(0.2\times10^{-3})}$$

$$2\theta = 5000 \times 10^{-6}$$

$$2\theta = 5 \times 10^{-3} \text{ radian}$$

The angular width of each fringe in Young's double slit experiment is given by: $\theta' = \dfrac{\lambda}{d}$.

Substitute the given values:

$$\theta' = \frac{(500\times10^{-9}\text{ m})}{(0.5\times10^{-3}\text{ m})}$$

$$\theta' = 1 \times 10^{-3} \text{ radian}$$

**(ii)** The number of fringes ($n$) that can be accommodated within the angular spread of the central maximum due to the single slit,

$$n = \frac{(2\theta)}{\theta'}$$

$$n = \frac{(5\times10^{-3})}{(1\times10^{-3})}$$

$$n = 5$$

**Concept:** Young's double slit experiment

**26.** **(i)** Heavy nuclei (like U-235) have lower binding energy per nucleon compared to elements like iron. Fission breaks these heavy nuclei into smaller, more stable fragments, which have higher binding energy per nucleon. This increase in stability releases the difference in binding energy as a tremendous amount of energy.

**(ii)** Light nuclei (like hydrogen isotopes) have a relatively low binding energy per nucleon and are less stable than heavier, fused products (like helium). Fusing these light nuclei together results in a nucleus with a higher binding energy per nucleon, thus releasing energy. However, positively charged light nuclei strongly repel each other (Coulomb barrier). To overcome this repulsion and allow the short-range strong nuclear force to bind them, extremely high temperatures (and pressures) are needed to give the nuclei enough kinetic energy to get close enough. This Coulomb barrier is lowest for light nuclei, making fusion more feasible with them.

**Concept:** Nuclei

**27.** **(I) (a)** The magnitude of induced emf,

$$|\varepsilon| = \frac{d\phi}{dt} = \frac{d}{dt}(BA) = B\frac{dA}{dt}$$

$$= B\frac{dA}{dt} = B\frac{d}{dt}(lx) = Bl\frac{dx}{dt} = Blv$$

**(b)** External force required to move the arm CD,

$$F = IlB = \frac{|E|}{R}lB = \frac{Blv}{R}\cdot lB = \frac{B^2l^2v}{R}$$

**(c)** Power dissipated as heat,

$$P = Fv = \frac{B^2l^2v}{R}\cdot v = \frac{B^2l^2v^2}{R}$$

**Concept:** Induced emf

**OR**

**(II) (a)** The magnetic flux ($\varphi$) linked with the coil at any instant t is given by:

$$\phi = NBA\cos(\theta)$$

$$\phi = NBA\cos(\omega t)$$

**(b)** According to Faraday's Law of Electromagnetic Induction,

$$\varepsilon = -\frac{d\phi}{dt}$$

Substitute the expression for $\phi$

$$\varepsilon = -\frac{d}{dt}[NBA\cos(\omega t)]$$

$$\varepsilon = -NBA\frac{d}{dt}\cos(\omega t)$$

$$\varepsilon = -NBA[-\omega\sin(\omega t)]$$

Therefore, the instantaneous induced electromotive force ($\varepsilon$) is:

$$\varepsilon = NBA\omega\sin(\omega t)$$

**(c)** The maximum induced emf ($\varepsilon_{max}$) occurs when $\sin(\omega t) = 1$.

So, $\varepsilon_{max} = NBA\omega$

Now the frequency ($f$) $= \dfrac{\text{Number of revolutions}}{\text{Time}}$

$$= \frac{10\text{ rev}}{2s} = 5\text{ Hz}$$

Angular speed ($\omega$) $= 2\pi f$

$$\omega = 2\pi(5\text{ Hz}) = 10\pi\text{ rad s}^{-1}$$

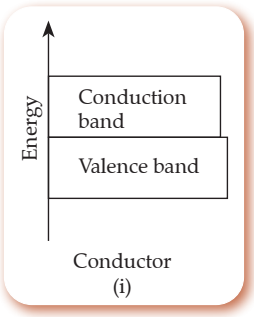
Therefore, $\varepsilon_{max} = (100)\times(0.2)\times(0.05)\times(10)$

$$= 10\pi\text{ V}$$

$$\varepsilon_{max} = 10 \times 3.14 = 31.4\text{ V}$$

**Concept:** Induced emf

**28.**

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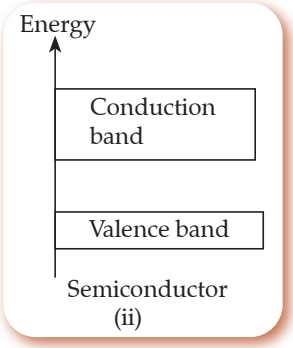


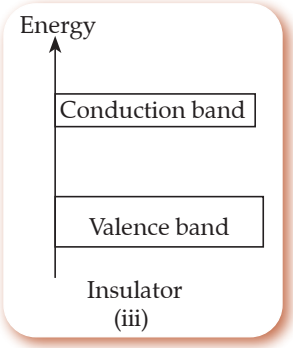


- Conductors have overlapping valence and conduction bands, allowing easy electron flow and high conductivity.
- Semiconductors have a small, energy gap between valence and conduction bands, allowing some electrons to cross with thermal energy for moderate conductivity.
- Insulators posseses a large energy gap, which prevents electrons from easily jumping to conduction band, resulting in every low conductivity

**Concept:** Semiconductor

## SECTION D

**29.** **(I) (a)** 50

**Why?** The dielectric constant ($k$) is defined as the ratio of the capacitance with the dielectric medium ($C$) to the capacitance without it ($C_0$).

$$k = \frac{C}{C_0} = \frac{200}{4} = 50$$

**(II) (d)** 80 pF

**Why?** The capacitance of a parallel plate capacitor is given by $C = \frac{\varepsilon_0 A}{d}$ for air, and $C' = \frac{K\varepsilon_0 A}{d'}$ with a dielectric.

Given $C_0 = 8\text{pF}$.

The new separation $d' = \frac{d}{2}$.

The new dielectric constant $k = 5$.

New capacitance

$$C' = \frac{k\varepsilon_0 A}{d'} = \frac{k\varepsilon_0 A}{(d/2)} = 2k\left(\frac{\varepsilon_0 A}{d}\right) = 2KC_0$$

$$C' = 2 \times 5 \times 8\text{pF} = 80\text{pF}.$$

**(III) (a)** 1:2

**Why?** Capacitance of the capacitor before inserting metal plate is $C = \frac{(\varepsilon_0 A)}{d}$ and after the insertion of any dielectric plate filling the capacitor partially is

$$C = \frac{(\varepsilon_0 A)}{\left[(d-t) + \frac{t}{K}\right]}$$

We know that $k$ for a metal is infinite thus $C = \frac{(\varepsilon_0 A)}{d-t}$

Here $t = \frac{d}{2}$ thus $C' = \frac{2\varepsilon_0 A}{d} = 2C$

$C:C' = 1:2$

**(IV) (d)** The potential difference between the plates decreases.

**Why?** For a charged, isolated parallel plate capacitor, when a dielectric is inserted, the potential difference between the plates decreases. The charge on the plates remains constant because it's an isolated system. The capacitance increases due to the dielectric material.

**Concept:** Capacitance

**30.** **(i)** When a charged particle enters a magnetic field perpendicularly, the magnetic force provides the necessary centripetal force for circular motion, i.e.,

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB}$$

**(ii)** The period of revolution is the time taken to complete one full circle.

So, $$T = \frac{2\pi r}{v}$$

Substitute the expression for $r$

$$T = \frac{2\pi\left(\frac{mv}{qB}\right)}{v}$$

$$T = \frac{2\pi m}{qB}$$

**(iii)** Radius $r = \frac{mv}{qB}$

For both the particles speed, magnetic field and charge are same.

So, $r \propto m$.

As mass of electron is less than mass of proton, therefore radius of the path followed by electron is smaller.

**Concept:** Moving charge in Magnetic field

## SECTION E

**31.** **(I) (a)** Bohr's model for hydrogen is based on three postulates:

- Stable Orbits: Electrons revolve in specific, non-radiating circular orbits where the electrostatic force provides the centripetal force.

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- Quantized Angular Momentum: Only those orbits are allowed where the electron's angular momentum is an integral multiple of h/(2π).
- Energy Transitions: Electrons emit or absorb energy only when transitioning between these fixed energy orbits, releasing or absorbing a photon whose energy equals the orbital energy difference.

Expression for Bohr radius:

Let us consider

$m$ = Mass of an electron

$r$ = Radius of the circular orbit in which the electron is revolving

$v$ = Speed of electron

$-e$ = Charge of electron

From 1st postulate

Centripetal force = Electrostatic force

$$\frac{mv^4}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}$$

$$\therefore \quad v^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{mr} \quad ...(i)$$

From 2nd postulate

Angular momentum is an integral multiple of $\frac{h}{2\pi}$

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr}$$

$$v^2 = \frac{n^2h^2}{4\pi^2m^2r^2} \quad ...(ii)$$

On comparing equations (i) and (ii),

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{mr} = \frac{n^2h^2}{4\pi^2m^2r^2}$$

$\therefore$ Bohr radius, $r = \frac{\varepsilon_0 n^2h^2}{\pi me^2}$

**(ii)** The shortest wavelength in Balmer series can be found using the expression:

$$\frac{1}{\lambda_S} = R\left(\frac{1}{2^2} - \frac{1}{\infty}\right)$$

$$\therefore \quad \lambda_S = \frac{4}{R}$$

Longest wavelength in Balmer series can be found using the expression:

$$\frac{1}{\lambda_L} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\therefore \quad \lambda_L = \frac{36}{5R}$$

So,

$$\frac{\lambda_L}{\lambda_S} = \frac{\frac{36}{5R}}{\frac{4}{R}} = \frac{9}{5}$$

**Concept:** Bohr's model of atom

**OR**

**(II) (a)** $mvr = \frac{nh}{2\pi}$ ...(i)

$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} \quad ...(ii)$$

(Since, Centripetal force = Electrostatic force)

On comparing equations (i) and (ii), we get

Bohr' radius, $r = \frac{\varepsilon_0 n^2h^2}{\pi me^2}$

Potential energy, $U = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r}$

$$= -\frac{me^4}{4\varepsilon_0^2 n^2h^2}$$

Kinetic energy,

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{nh}{2\pi mr}\right)^2$$

$$= \frac{n^2h^2\pi^2m^2e^4}{8\pi^2m\varepsilon_0^2n^4h^4}$$

$$KE = \frac{me^4}{8\varepsilon_0^2n^2h^2}$$

Total energy = K.E. + P.E.

$$= -\frac{me^4}{8\varepsilon_0^2n^2h^2} + \frac{-me^4}{4\varepsilon_0^2n^2h^2}$$

$$= -\frac{me^4}{8\varepsilon_0^2n^2h^2}$$

**(b) Rydberg's formula:** For first member of Lyman series,

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$

$$\lambda = \frac{4}{3R} = \frac{4}{3}\times 912\text{Å}$$

$$= 1216\text{Å}$$

For first member of Balmer Series,

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\lambda = \frac{36}{5R}$$

$$= \frac{36}{5}\times 912\text{Å}$$

$$= 6566.4\text{Å}$$

**Concept:** Bohr's model of atom

**32.** **(I) (a)** Expression for the equivalent magnetic moment of a planar current loop:

The equivalent magnetic moment $\vec{m}$ of a planar current loop with $N$ turns, area $A$ and carrying a current $I$ is given by:

$$m = NIA$$

As a vector, $\vec{m} = NI\vec{A}$, where $\vec{A}$ is the area vector perpendicular to the plane of the loop

Magnetic dipole moment of a revolving electron:

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Consider an electron of charge $e$ revolving in a circular orbit of radius $r$ with a speed $v$. This motion constitutes a current loop.

The current ($I$) due to the revolving electron

$$I = \frac{e}{T}$$

The time period of revolution, $T = \frac{2\pi r}{v}$

Substituting $T$ into the current expression:

$$I = \frac{e}{\frac{2\pi r}{v}}$$

$$= \frac{ev}{2\pi r}$$

The area of the electron's orbit is $A = \pi r^2$.

For a single electron orbit, $N = 1$.

Therefore, the magnetic dipole moment ($\mu_1$) due to the orbital motion of the electron is:

$$\mu_1 = NIA = (1)\times\left(\frac{ev}{2\pi r}\right)\times(\pi r^2)$$

$$\mu_1 = \frac{evr}{2}$$

**(b)** The magnetic potential energy ($U$) of a magnetic dipole in a uniform magnetic field $(\vec{B})$ is given by

$$U = -\vec{m}\cdot\vec{B} = -mB\cos\theta$$

The circular loop has radius $r$, $N$ turns, and carries current $I$. It is kept in the $XY$ plane.

The area of the loop is $A = \pi r^2$.

Since the loop is in the $XY$ plane, its area vector ($\vec{A}$) is perpendicular to this plane, i.e., along the $Z$-axis.

Therefore, $\vec{A} = (\pi r^2)\hat{k}$.

The magnetic moment of the coil $m = NIA$.

So, $\vec{m} = NI(\pi r^2)\hat{k}$

The uniform magnetic field is given as

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}.$$

Substitute the expressions for $\vec{m}$ and $\vec{B}$ into the potential energy formula:

$$U = -\vec{m}\cdot\vec{B}$$

$$U = -[NI(\pi r^2)\hat{k}]\cdot[\vec{B}_x\hat{i} + \vec{B}_y\hat{j} + \vec{B}_z\hat{k}]$$

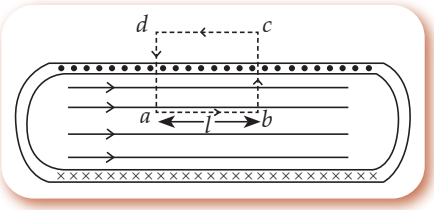
$$U = -NI(\pi r^2)\,B_z$$

**Concept:** Magnetic moment

**OR**

**(II) (a) Ideal Solenoid:** For a very long solenoid, the magnetic field inside is uniform and parallel to the axis, and the magnetic field outside is negligible.

**Amperian Loop:** We construct a rectangular Amperian loop, abcd, with its length ($l$) parallel to the solenoid's axis, inside the solenoid. The other side of the loop extends outside the solenoid.




We now evaluate the line integral of $\vec{B}\cdot d\vec{l}$ over each segment of the rectangular loop:

- Along segment ab (inside the solenoid):

$\int_a^b \vec{B}\cdot d\vec{l} = Bl$ (since $\vec{B}$ is parallel to $d\vec{l}$).

- Along segments bc and da (perpendicular to the axis inside the solenoid):

$\int_b^c \vec{B}\cdot d\vec{l} = 0$ and $\int_d^a \vec{B}\cdot d\vec{l} = 0$ (since $\vec{B}$ is perpendicular to $d\vec{l}$).

- Along segment $cd$ (outside the solenoid): $\int_c^d \vec{B}\cdot d\vec{l} = 0$ (since $\vec{B} \approx 0$ outside).

Total Line Integral: Summing these up, $\oint \vec{B}\cdot d\vec{l} = Bl$.

If there are $n$ turns per unit length, then the number of turns in length $l$ is $nl$.

Therefore, the total current enclosed by the loop is $I_{enc} = nIl$.

Applying Ampere's law, $Bl = \mu_0\, nIl$.

This simplifies to $B = \mu_0\, nI$.

**(b) (i)** $\chi < 0$ (Diamagnetic Material):

- Diamagnetic materials are weakly repelled by magnetic fields.
- When suspended freely in Earth's magnetic field, they align themselves perpendicular to the field.

**(ii)** $\chi >> 1000$ (Ferromagnetic Material):

- Ferromagnetic materials are strongly attracted to magnetic fields and can retain their magnetism.
- When placed in Earth's magnetic field, they will experience a strong force that aligns their magnetic domains with the direction of the field.
- Therefore, the bar will orient itself such that its length aligns with the Earth's magnetic field.

**Concept:** Solenoid

**33.** **(I) (a)** Set A: stable interference pattern, the positions of maxima and minima do not change with time.

Set B: positions of maxima and minima will change rapidly with time and an average uniform intensity distribution will be observed on the screen.

**(b)** Expression for intensity of stable interference pattern in set –A

If the displacement produced by slit $S_1$ is

$$y_1 = a\cos\omega t$$

then, the displacement produced by $S_2$ would be

$$y_2 = a\cos(\omega t + \varphi)$$

and the resultant displacement will be given by

$$y = y_1 + y_2$$

$$= a[\cos\omega t + \cos(\omega t + \varphi)]$$

$$= 2a\cos\left(\frac{\phi}{2}\right)\cos\left(\omega t + \frac{\phi}{2}\right)$$

The amplitude of the resultant displacement is $2a\cos\left(\frac{\phi}{2}\right)$ and therefore the intensity at that point will be

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$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

$$\phi = 0$$

$$\therefore \quad I = 4I_0$$

In set $B$, the intensity will be given by the average intensity

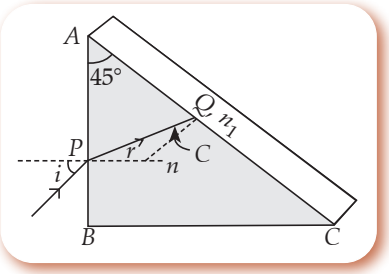
$$(I) = 4I_0\left(\cos^2\left(\frac{\phi}{2}\right)\right)$$

$$I = 2I_0$$

**Concept:** Interference of light

**OR**

**(II) (a)**




Given, $(n_1 < n)$

$$n \sin C = n_1 \sin 90°$$

$$\sin C = \frac{n_1}{n}$$

In $\Delta APQ$, $90° - r + 90° - C + 45° = 180°$

$$r + C = 45°$$

$$\Rightarrow \quad r = 45° - C$$

Considering refraction at $AB$

$$1 \times \sin i = n \sin r$$

$$= n \sin (45° - C)$$

$$= n \sin 45° \cos C - \cos 45° \sin C$$

$$= n\left[\frac{1}{\sqrt{2}}\sqrt{1-\frac{n_1^2}{n^2}} - \frac{1}{\sqrt{2}}\frac{n_1}{n_2}\right]$$

$$= \frac{1}{\sqrt{2}}\left[\sqrt{n^2 - n_1^2} - n_1\right]$$

$$i = \sin^{-1}\left[\frac{1}{\sqrt{2}}\{\sqrt{n^2 - n_1^2} - n_1\}\right]$$

**(b)** If refracted ray passes undeviated through AC, for this ray must be incident normally on AC i.e., $r' = 0$

$$r + r' = A \Rightarrow r = 45°$$

$$1 \times \sin i = n \sin 45° = \frac{1.352}{\sqrt{2}}$$

$$i = \sin^{-1}\left(\frac{1.352}{\sqrt{2}}\right) = \sin^{-1}(0.96)$$

**Concept:** Refraction through Prism

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